

Package ‘ASEXtras4’

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Type Package

Title Utility functions for asreml objects

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Author David Butler

Maintainer <david.butler@uow.edu.au>

Description Diagnostic functions for the meta-analysis of a series of spatially defined field experiments.

Depends R (>= 3.5.0)

Imports asreml, ggplot2, data.table (>= 1.9.6), cluster, lattice

License GPL

LazyLoad yes

RoxygenNote 7.1.0

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fa.asreml	<i>Factor analytic variance models.</i>
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Description

Summarises an asreml model in which factor analytic (FA) variance structures have been fitted. The function aids in interpreting FA models fitted to genotype by environment effects in the analysis of plant breeding multi-environment trial (MET) data.

Usage

```
fa.asreml(
  object,
  uniplot = TRUE,
  uniplot.tol = 0.85,
  uniplot.cex = 0.5,
  trunc.char = c(1, 6),
  blups = TRUE,
  regplot = TRUE,
  addedplot = TRUE,
  g.list = NULL,
  heatmap = TRUE,
  heatmap.ord = "asis",
  agnes.method = "average"
)
```

Arguments

<code>object</code>	An asreml object.
<code>uniplot</code>	if TRUE (the default), pairwise plots of FA loadings are produced for $k > 1$.
<code>uniplot.tol</code>	A numeric value between 0 and 1 that affects the display of pairs of loadings in the uniplots. Those pairs with a (scaled) euclidean distance greater than <code>uniplot.tol</code> are drawn with solid blue lines, otherwise they appear as red dashed lines.
<code>uniplot.cex</code>	Character expansion (relative) size for plotting (the default is 0.5).
<code>trunc.char</code>	A numeric vector of length 2 specifying the truncation of loading labels for use in plotting. The default (<code>c(1, 6)</code>) is to use the first 6 characters of the environment factor levels, otherwise if NULL the complete level names are used.
<code>blups</code>	If TRUE (the default), summaries of estimates from the asreml object are produced. These include BLUPs of genotype by environment effects and genotype scores, and REML estimates of FA variance parameters.
<code>regplot</code>	If TRUE (the default), regression plots of genotype by environment effects against FA loadings are produced.
<code>addedplot</code>	If TRUE (the default), added variable plots of genotype by environment effects against FA loadings are produced.
<code>g.list</code>	A character vector comprising the subset of genotype levels to be used in the regression and added variable plots. The default (NULL) is to use the complete set of genotype levels.
<code>heatmap</code>	If TRUE (the default), a heatmap of the estimated FA correlation matrix is produced.
<code>heatmap.ord</code>	A character string ("asis", "cluster") or vector of environment levels in the order desired for plotting the heatmap. The default is "asis", where the rows and columns of the correlation matrix are maintained in environment level order. The "cluster" option invokes the agnes clustering algorithm on the FA correlation matrix then orders the rows and columns accordingly.
<code>agnes.method</code>	Clustering strategy; the default is "average".

Details

In the analysis of MET data, the FA model is applied to the vector of random genotype by environment effects (*Smith et al, 2001*). The model can be thought of as a (multiple) regression of genotype by environment effects on environment covariates (also called *loadings*) with separate slopes (*scores*) for each genotype. We denote the number of terms or *factors* in the model by k , the number of environments by t and varieties by m .

The variance parameters associated with this FA model are the loadings (t_k of these) and the so-called specific variances (t of these) which can be regarded as the regression lack-of-fit or residual variances for individual environments. Together the loadings and specific variances provide a form for the genetic variance matrix for environments (diagonal elements are the genetic variances for individual environments; off-diagonal elements are the genetic covariances between pairs of environments). `fa.asreml` returns the REML estimate of the genetic variance matrix (additionally converted to a correlation matrix). A graphical depiction of the correlation matrix as a heatmap is optionally available. If requested the rows and columns on the heatmap may be re-ordered on the basis of the application of the *agnes* clustering algorithm to the genetic correlation matrix (See *Cullis et al, 2010*).

`fa.asreml` also returns REML estimates of the loadings and specific variances. The loadings have been rotated to a principal component solution to aid interpretation. The loadings may be depicted graphically as *uniplots* in which pairs of factors are plotted against each other. The pair of loadings for an individual environment is represented as a line from the origin. To assist interpretation, the loadings are first converted to the correlation rather than covariance scale. The unit circle is imposed on each uniplot as it reflects the upper limit where 100% of genetic variance is explained by the two factors being plotted (See *Cullis et al, 2010*).

`fa.asreml` also returns BLUPs of the genotype effects for each environment (t_m of these) and BLUPs of the genotype scores for each factor (k_m of these).

Plots that reflect the multiple regression implicit in the FA model are also optionally available. These may either be regression or added variable plots. With the former, the genotype by environment BLUPs are plotted against the loadings, with a separate panel for each factor and each (pre-specified) genotype. The BLUPs are plotted as points. Super-imposed on each panel is the regression line, that is, with slope given by the genotype score for that factor. The added variable plots are similar except that the points on the panels for factor $i, i = 2 \dots k$ are the regression residuals associated with the first $i - 1$ factors.

Note that several FA models are allowed within the `asreml` call. For example if pedigree information is included in the analysis there may be separate FA models (possibly of different order) for the additive and non-additive genotype by environment effects (*Cullis et al, 2010*). We let f denote the number of FA terms in the model (typically $f \leq 2$). Also, the factor names *Environment* and *Genotype* are used generically here, however, `fa.asreml` uses the factor names appearing in the FA term(s) in the `asreml` model.

Value

A list with the following components:

blups a list of length f . Each element is itself a list, with elements:

blups A data-frame of length m_t containing the BLUPs of the genotype effects for each environment and the associated regression BLUPs (predictions from the FA regression terms). These two differ in that the latter does not include the regression residuals. Also given is the number of non-missing y -values (may be zero) for each genotype in each environment.

scores A data-frame of length m_k containing the BLUPs of the genotype scores for each factor.

gammas A list of length f where each element is a list with components:

Gmat The REML estimate of the genetic variance matrix (dimension $t \times t$)

Cmat The REML estimate of the genetic correlation matrix (dimension $t \times t$)

rotated loads A matrix of dimension $t \times k$ containing the REML estimates of the FA loadings, rotated to a principal component solution.

rotated loads - c A matrix of dimension $t \times k$ containing the REML estimates of the FA loadings, rotated to a principal component solution and re-scaled to align with Cmat rather than Gmat.

specific var A vector of length t containing the REML estimates of the specific variances.

site %vaf A matrix of dimension $t \times k$ containing, for each environment, the percentage of genetic variance accounted for by the FA regression, both as a whole and by individual factors.

total %vaf A scalar giving the total percentage of genetic variance accounted for by the FA regression.

uniplots A list of length f . Each element may be a trellis object containing a pairwise plot of factor loadings 1 and 2 ($k = 2$), or a list of $k(k - 1)/2$ trellis objects ($k > 2$), one element for each pairwise plot.

regplots A list of length f . Each element is a trellis object containing plots of the genotype by environment BLUPs against factor loadings. There is a separate panel for each factor and genotype specified in `g.list`. Superimposed on each panel is the associated regression line with slope given by the genotype score for that factor.

addedplots A list of length f . Each element is a trellis object similar to that in `regplots` except it contains added variable plots. The points on the panels for factor i , $i = 2 \dots k$ are the regression residuals associated with the first $i - 1$ factors. When $k = 1$ the graphics in `regplots` and `addedplots` are identical.

heatmap A list of length f . Each element is a trellis plot containing a heatmap of the genetic correlation matrix. The rows and columns will be ordered as per the `heatmap.ord` argument.

agnes A list of length f where each element is the standard output from the `agnes` function. Returned only if the `heatmap.ord` argument was specified with the "cluster" option.

References

Cullis BR, Smith AB, Beek C, Cowling WA (2010). "Analysis of Yield and Oil from a Series of Canola Breeding Trials. Part II: Exploring VxE using Factor Analysis." *Genome*, **53**, 1002-1016.

Smith AB, Cullis BR, Thompson R (2001). "Analyzing Variety by Environment Data Using Multiplicative Mixed Models and Adjustments for Spatial Field Trend." *Biometrics*, **57**, 1138-1147.

faces.asreml

Variogram faces

Description

Computes and plots the approximate coverage intervals for the faces of the semi-variogram of the residuals from a spatially defined experiment.

Usage

```
faces.asreml(
  obj,
  sim = 100,
  trace = FALSE,
  trim = 20,
  main = "Approximate 95% coverage interval"
)
```

Arguments

<code>obj</code>	An asreml object.
<code>sim</code>	The number of simulations, default is 100.
<code>trace</code>	If TRUE, asreml convergence monitoring is written to the R console (the default is FALSE).
<code>trim</code>	Variogram values based on less than <code>trim</code> points are discarded (the default is 20).
<code>main</code>	Title for the convergence plots.

Details

For each of `sim` iterations, a *new* dataframe is constructed from sampling the distributions of the random model terms. Only single section models with simple variance parameters in the random formula are allowed.

Value

A list with the following components:

plots A list of length two containing the trellis plots for the variogram faces.

vparameters A two-column matrix of the fitted and (mean) simulated random components.

tau A two-column matrix of the fitted and (mean) simulated fixed effects.

variogram The data frame of variogram summaries used in plots.

References

Stefanova KT, Smith AB, Cullis BR (2009). "Enhanced Diagnostics for the Spatial Analysis of Field Trials." *Journal of Agricultural, Biological, and Environmental Statistics*, **14**(4), 392-410.

met.asreml

Multi-environment trials.

Description

Plot the sample variogram and spatial trends in residuals from the meta-analysis of a series of field experiments.

Usage

```
met.asreml(
  object,
  plottype = "together",
  type = "default",
  pages = NA,
  trim = 6,
  spatial = "plot",
  npanels = NA,
  swap.axes = TRUE,
  new.device = FALSE,
  display = TRUE,
  ...
)
```

Arguments

<code>object</code>	An object of class <code>asreml</code> , or class <code>met.asreml</code> from a previous call.
<code>plottype</code>	A character string ("together", "separate") specifying the layout of the plots. The default "together" draws a trellis display of residuals against the spatial dimensions, and a variogram of the residuals in the one graphics frame. Otherwise, the plots appear in separate frames.
<code>type</code>	Type of residuals, see <code>residuals.asreml</code> .
<code>pages</code>	A numeric or character vector identifying the pages to display from a previous call to <code>met.asreml</code> . If NA (the default), all plots are (re)displayed.
<code>trim</code>	An integer specifying the minimum number of observations for a point to be included in the semi-variogram; the default is 6.
<code>spatial</code>	Whether to include a nugget effect; see <code>residuals.asreml</code> .
<code>npanels</code>	If <code>plottype == "separate"</code> , <code>npanels</code> specifies the number of trellis panels per page. If NA (the default), a suitable default is calculated.
<code>swap.axes</code>	If TRUE (the default), the dimension with the highest number of levels is on the x-axis, otherwise the order specified in the residual formula is retained.
<code>new.device</code>	if TRUE (default is FALSE), a new trellis device is opened for each plot. Ignored if <code>plottype == "separate"</code> .
<code>display</code>	If TRUE (the default), render the plots on the default display. <code>met.asreml</code> silently returns a list of the trellis objects produced.
<code>...</code>	Additional arguments to be passed to the lattice functions.

Details

`met.asreml` displays the residuals from a series of one or more experiments in a set of "pairs" of trellis plots. Each pair contains a trellis plot of the residuals against the factors defining the *dimensions* in the residual formula of an `asreml` model, and an empirical semi-variogram of the residuals against these factors. Each pair may appear in a single trellis frame (`plottype="together"`), or in separate trellis devices.

Value

An object of class `met.asreml` containing a list of trellis objects, with elements named by the levels of the factor defining the *sections* in the residual model formula.

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